

GALLANT IAS ACADEMY
PRELIMS CUM MAINS BATCH 2027
CURRENT AFFAIRS DECODED
SUMMARY NOTES
11th July to 18th July, 2026

Article : Deadly Parrot Virus PaBV-4 Found in Indian Pet Parrots

Indian scientists have, for the first time, found **PaBV-4 (Parrot Bornavirus 4)**, a deadly virus, in pet parrots kept in cages in India. Infected birds ranged from near-threatened to endangered, raising concern that the virus could wipe out rare, threatened parrot species.

- The finding exposes gaps in how India controls the health, breeding, and trade of exotic pet birds.

CITES

The **Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES)** is an international agreement between governments ensuring that international trade in wild animals and plants does not threaten their survival.

- Came into force in 1975; India joined in 1976. It now has 184+ member countries and is legally binding on all of them.
- **Appendix I**: species most threatened with extinction; commercial international trade is banned.
- **Appendix II**: species not yet threatened but that could become so without controlled trade; trade allowed with permits.
- **Appendix III**: species a country asks other members to help protect; India has listed species here, such as Red Sandalwood.

Article : India's First Hydrogen-Powered Train

On 17 July 2026, PM Modi is set to launch India's first hydrogen-powered train, running between Jind and Sonipat in Haryana.

How a Hydrogen Fuel Cell Works

- A fuel cell makes electricity through a chemical reaction, taking in hydrogen gas stored on board and oxygen pulled from outside air.
- The only waste product is water; there is no smoke or carbon dioxide, which is why it is called **zero-emission technology**.
- The fuel cell always produces a constant amount of power; the onboard battery, by contrast, can either take in or give out power depending on what is needed at that moment.

Global Precedent

- French manufacturer **Alstom** first showed off hydrogen train technology at a 2016 exhibition in Berlin.
- In 2018, its two-car hydrogen train, the **Coradia iLint**, began running in Germany, becoming the world's first hydrogen-powered passenger train.

Types of Hydrogen (by Production Method)

Colour	Source
Green	Water split by renewable energy
Blue	Fossil fuel with emissions captured
Grey	Natural gas
Brown	Lignite coal
Black	Bituminous coal
Turquoise	Methane
Pink / Purple	Water split using nuclear power and heat
White	Mined directly from underground reservoirs
Orange	Made from recycled plastic waste

National Green Hydrogen Mission

- Targets 5 MMT annual green hydrogen production and 125 GW of renewable capacity by 2030.
- **SIGHT Scheme:** provides financial incentives for domestic electrolyser manufacturing and green hydrogen production.
- Developing dedicated **Green Hydrogen Hubs** near major industrial clusters and ports for seamless export.

Article : INS Mahendragiri Commissioned into the Navy

On 11 July 2026, the Indian Navy commissioned **INS Mahendragiri** (pennant number **F38**) at the Naval Dockyard, Visakhapatnam, Andhra Pradesh, the sixth of seven Project 17A stealth frigates to be commissioned. The ship joins the Eastern Fleet (the "Sunrise Fleet"), based at Visakhapatnam.

Warship Classes

- **Corvette:** the smallest ocean-going warship, usually for coastal patrolling.
- **Frigate:** a medium, multi-role ship.
- **Destroyer:** larger and more heavily armed, built for fleet air defence and offensive strike.
- **Aircraft carrier:** a floating air base that carries fighter jets.

Project 17A

Project 17A (P17A) is an Indian Navy programme to build seven advanced stealth frigates, the **Nilgiri class**; "17A" denotes the improved "Alpha" version of the earlier Project 17 (Shivalik class).

- The seven Nilgiri-class ships: INS Nilgiri, Udaygiri, Himgiri, Taragiri, Dunagiri, Mahendragiri, and Vindhyagiri.

Features of INS Mahendragiri

- Can be equipped with the **BrahMos** surface-to-surface missile.
- Carries an indigenous rocket launcher and torpedo launchers.
- Fitted with an Integrated Anti-Submarine Defence System.

Article : H-1B Visas

The **H-1B visa** is a non-immigrant, employer-sponsored US work visa created under the Immigration Act, 1990, allowing US employers to temporarily hire foreign professionals for jobs requiring a high level of skill.

- Helps fill skill gaps in sectors such as IT, engineering, finance, and healthcare.
- Visas are capped each year and allocated through a lottery run by the **US Citizenship and Immigration Services (USCIS)**.
- Validity: initially 3 years, extendable up to 6 years.
- People born in India have been the largest beneficiaries, accounting for over 70% of all approved H-1B petitions annually since 2015.

Article : AI Use by the Judiciary - What SC's Proposed Regulations Allow

The Supreme Court released the draft **Regulations for Use of Artificial Intelligence (AI) in Courts, 2026**, aiming to build a governance framework for AI in the judiciary, with public comments sought till 15 July.

- The Regulations are not automatically binding: they come into force for the SC on a date notified by the CJI, while High Courts and subordinate courts adopt them separately on dates set by each High Court's Chief Justice.
- **Permitted uses:** administrative and assistive functions such as case management, transcription, translation, legal research, document summarisation, accessibility, and court administration.
- **Prohibited uses:** AI cannot decide a case; it may not be used for risk scoring of flight risk, predicting recidivism, evaluating bail eligibility, judging witness credibility, or applying black-box AI in matters affecting personal liberty.
- **Institutional architecture:** an "Apex Body" at the SC (sitting judges, a MeitY official, finance and cybersecurity experts) leads regulation; each court gets its own AI Committee, backed by a new **Centre of Research and Excellence on AI (CoRE-AI)**.

AI Terms Glossary

- **Weak AI / Narrow AI:** limited to a specific area, simulating human cognition.
- **Strong AI:** systems carrying out human-like tasks, more complex than narrow AI.
- **Augmented Intelligence:** systems designed to work alongside humans to improve task performance.
- **Theory-of-mind AI:** fully adaptive systems with an extensive ability to learn and retain past experience, such as advanced chatbots that could pass the **Turing Test** (a method for judging whether a computer can think like a human, by fooling a person into believing it is human).
- **Black Box AI:** AI whose decision-making process is difficult or impossible for humans to understand.
- **Explainable AI (XAI):** AI that provides reasons or explanations for its decisions.
- **Agentic AI:** AI that can plan, decide, and perform tasks independently to achieve

a goal.

- **Generative AI:** AI that creates new content such as text, images, music, video, or code.
- **AI Hallucination:** when AI generates false or fabricated information while presenting it as true.
- **Data Mining:** extracting meaningful patterns or knowledge from large datasets.

Article : Bishnoi Indictment - The Long Road to Extradition

Days after indicting gangster **Lawrence Bishnoi** in a series of crimes including the killing of Canada-based separatist Hardeep Singh Nijjar, the US Department of Justice confirmed it will seek his extradition.

Legal Framework

- Extradition between India and the US is governed by the bilateral **Extradition Treaty (1997)** and India's **Extradition Act, 1962**; central to the treaty is "dual criminality": the offence must be punishable by over one year's imprisonment in both countries.
- Even if treaty conditions are met, extradition is not automatic: **Section 31** of the Extradition Act allows India to postpone surrender while domestic proceedings are pending.
- **The process:** a formal US request goes through the MEA, is examined against treaty and Act requirements, certified as extraditable by an Indian court, with the final surrender decision resting with the Union government.
- Extraditions between India and the US are infrequent: only 11 fugitives were extradited by the US to India between 2002-2018, against 60 pending Indian requests in that period.

Key Provisions of the Extradition Act, 1962

- Covers both treaty-based and non-treaty extradition cases.
- **Dual criminality principle:** the offence must be a crime in both India and the requesting country.
- Extraditable offences include terrorism, organised crime, drug trafficking, and financial fraud.
- **Bar on political offences:** individuals cannot be extradited for political offences.
- **Rule of specialty:** the fugitive can only be tried for the crime for which they were extradited.
- **Human rights considerations:** no extradition where there is a risk of torture or an unfair trial.

Nodal Authorities

- **Ministry of External Affairs (MEA):** processes extradition requests from foreign governments.
- **Central Bureau of Investigation (CBI):** handles extradition-related investigations.
- **National Investigation Agency (NIA):** pursues cases related to terrorism and national security.

Article : Five Crore Indians Wait When the Courts Take a Break

As of end-2025, over 5.39 crore cases were pending in Indian courts (4.76 crore in district courts, 63.6 lakh in High Courts, and over 92,000 in the Supreme Court, its

heaviest load in over 30 years).

- The Supreme Court and major High Courts take a six-week-plus summer break annually, with only limited vacation benches functioning; the practice traces back to English judges retreating from Indian heat.
- In 2024, the Court renamed "summer vacation" as "partial court working days," but the actual number of sitting days (about 190 a year) remained unchanged, a rebranding rather than a structural fix.
- A 2023 parliamentary standing committee, the Law Commission, and the Justice Malimath Committee have all proposed **staggered leave**, rotating leave among judges rather than the entire court going on vacation en masse.

Article : China's Long March in Space Tech Continues With Reusable Rocket

China successfully landed a reusable rocket for the first time; the **Long March 10B**, capable of carrying up to 16 metric tonnes to Low Earth Orbit, lifted off from the Wenchang Commercial Space Launch Site in Hainan.

India's Equivalent Effort

ISRO's **Reusable Launch Vehicle - Technology Demonstrator (RLV-TD)**, named **Pushpak**, is a winged technology demonstrator that simulates an aircraft-like landing. ISRO has successfully conducted the LEX (Landing Experiment) series (LEX-01, 02, 03) using Pushpak to test autonomous horizontal landing.

Article : Can Biogas Aid India's Energy Security?

What Is Biogas?

Biogas is a renewable fuel created by breaking down organic waste without oxygen (anaerobic decomposition). Microorganisms decompose the material inside an airtight tank, producing a methane-rich gas used for heat and electricity. Aerobic processes require oxygen to produce energy, while anaerobic processes produce energy without oxygen.

Compressed Biogas (CBG)

Raw biogas from waste cannot be used in vehicles right away; it is dirty, weak, and contains too many impurities. Turning it into CBG needs two steps: **upgrading** (stripping away impurities) and **compression** (packing it safely into metal cylinders for use in vehicles).

- CBG is chemically identical to CNG, so it can be used anywhere CNG is used, without new engines or pipelines.
- Biogas counts as green because the CO₂ released when it burns is roughly the same CO₂ that would have escaped from the rotting organic waste anyway, adding very little net new carbon, unlike fossil fuel; it is also made from waste, solving a waste-disposal problem while creating fuel.

Key Schemes

- **SATAT** (2018, Sustainable Alternative Towards Affordable Transportation): under the Ministry of Petroleum and Natural Gas, it set a goal of 5,000 CBG plants by 2023, producing 15 million tonnes of CBG a year; only 132 plants had actually been completed by 2026.
- **GOBARdhan** (2018, Galvanising Organic Bio-Agro Resources Dhan): launched under the Swachh Bharat Mission to convert cattle dung and farm/organic waste

into biogas; nodal agency is the Ministry of Jal Shakti.

Denmark's Model

- Denmark discourages using food crops as feedstock, with a cap limit; its biogas mainly comes from livestock manure and agricultural waste, avoiding the food-versus-fuel conflict Germany ran into.

Article : Vulnerabilities Found in the UMANG Portal

Researchers have recently revealed significant vulnerabilities in the **UMANG** portal.

- **UMANG (Unified Mobile Application for New-age Governance)** is a single unified platform to access government services offered by Central, State, and Local bodies, anytime, anywhere; it is part of the Digital India initiative of the Ministry of Electronics and Information Technology (MeitY).

Article : Supreme Court Constitutes Four Special Benches for Oldest Pending Cases

The Supreme Court has constituted four special benches to exclusively hear its oldest pending civil and criminal cases.

- Two division benches will exclusively hear the oldest civil cases, while two others will focus solely on long-pending criminal matters.
- The special benches will function on Tuesdays, Wednesdays, and Thursdays.
- Priority will go to decades-old disputes, including civil cases pending since 1986 and criminal matters dating back to 1991.

Article : Supreme Court Flags Growing Misuse of the POCSO Act

The Supreme Court has raised serious concerns about growing misuse of the **Protection of Children from Sexual Offences (POCSO) Act, 2012**, India's first comprehensive law focused solely on preventing and punishing sexual crimes against children.

- **Definition of child:** any person below the age of 18 years.
- **Gender neutrality:** protects both male and female children from any gender of offender.
- **Age of consent:** explicitly set at 18, criminalising all sexual activity below this age, regardless of consent between minors.
- **Confidentiality of identity:** forbids disclosure of the victim's identity in any media without court authorisation.
- **Time-bound trials:** mandates completion of investigation within two months and trial within one year of FIR registration.

Article : India-Vietnam Relations

Vietnam sits along the eastern coast of the Indochina Peninsula in Southeast Asia, sharing land borders with China to the north and Laos and Cambodia to the west, with a strategic coastline along the South China Sea.

- Politically, Vietnam is a single-party socialist republic where the Communist Party of Vietnam holds exclusive leadership, transformed economically by the 1986 Doi Moi reforms into an export-led market economy.
- Bilateral relations were elevated to a **Comprehensive Strategic Partnership** in 2016; Vietnam is a vital pillar of India's Act East Policy and a key partner in its

Indo-Pacific vision.

- India and Vietnam collaborate within regional frameworks including ASEAN and the Mekong-Ganga Cooperation.

Mekong-Ganga Cooperation (MGC)

Launched in 2000 in Vientiane, the MGC comprises six member countries: India, Cambodia, Laos, Myanmar, Thailand, and Vietnam, focused on economic, cultural, and educational ties, and closer collaboration in tourism, transport, and communications.

ASEAN

ASEAN's 11 member states are Indonesia, Malaysia, the Philippines, Singapore, Thailand, Brunei, Vietnam, Laos, Myanmar, Cambodia, and Timor-Leste, headquartered in Jakarta, Indonesia.

Article : In Census, Caste Category Must Not Be Open-Ended

The **Census** is a Union Subject under the Constitution.

Historical Context

- Between 1881-1931, caste data was regularly collected; 1931 was the last complete caste enumeration.
- Post-1951, independent India's censuses excluded caste enumeration, except for Scheduled Castes (SCs) and Scheduled Tribes (STs).
- The **SECC 2011** collected caste data, but the findings were not published due to concerns about data quality and consistency.
- A **survey** is conducted for a limited time and specific region, while a **census** is a statutory, nationwide process under the Constitution.

Anthropological Survey of India (AnSI)

AnSI is a government-funded organisation conducting anthropological research on India's diverse cultures, the only such research body in the Central Government, under the Ministry of Culture; headquartered in Kolkata.

- Established on 1 December 1945 under Dr S.C. Roy, initially to study the tribes and castes of India and their way of life; it originated as a Zoological and Anthropological Section under the Geological Survey of India in 1902, becoming the Anthropological Section of the Zoological Survey of India in 1916 before gaining independent status in 1945.

The Current Debate

- Reports indicate the rehearsal for Census 2027's second phase includes an "open column" for recording caste; the government says the final methodology will be decided after evaluating test-exercise feedback.
- The open-column approach in SECC 2011 produced over 46 lakh different caste entries due to spelling variations, regional names, and inconsistent reporting, data too messy to officially release.
- Structured state exercises worked better: **Bihar's Caste Survey (2023)** identified 209 castes, and **Telangana's caste survey (2024)** identified 242 castes, both using officially recognised caste lists with standardised codes.
- The proposed fix: the Union government should constitute an expert committee to determine an authoritative caste-coding source, drawing on state-wise lists, the central OBC list, and AnSI's caste database.
- The larger point: a well-designed caste census should generate accurate data to

identify deprivation and formulate targeted policies, which depends on a transparent, standardised, nationally accepted coding system.

Article : India-US Defence Tech Ties - Big Ambitions, Little Delivery

Key Agreements Timeline

Agreement	Year	What it does
DTTI (Defence Technology and Trade Initiative)	2012	Promotes co-development and co-production of defence technologies, moving beyond simple arms sales.
GSOMIA (General Security of Military Information Agreement)	2002	Allows militaries to share gathered intelligence; initially limited to public companies.
Major Defence Partner (MDP) status	2016	US designation facilitating access to advanced defence technology and recognising India's strategic importance.
LEMOA (Logistics Exchange Memorandum of Agreement)	2016	Allows access to each other's designated military facilities for refuelling and replenishment; not binding, needs clearance per request.
ISA (Industrial Security Annex)	2019	Extension of GSOMIA signed at the 2+2 dialogue, extending exchange of classified military information to private defence companies too.
COMCASA (Communications and Information Security Memorandum of Agreement)	2018	Legal framework for transferring sensitive communication security equipment, valid for 10 years, easing interoperability.
2+2 Ministerial Dialogue	2018	Annual dialogue between Defence and Foreign Ministers of both countries; the highest institutional defence dialogue.
BECA (Basic Exchange and Cooperation Agreement)	2020	Covers geospatial intelligence and sharing of maps and satellite images for defence.
INDUS-X (India-US Defense Acceleration Ecosystem)	2023	Fosters defence innovation and industrial cooperation.

SOSA (Security of Supply Arrangement)	2023	Legally non-binding; secures mutual priority delivery of defence goods during peacetime, emergency, or conflict; India is the 18th SOSA partner.
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The Core Critique

- India-US defence cooperation has generated ambitious political announcements that repeatedly fail to translate into industrial outcomes, despite the relationship expanding considerably in dollar terms.
- **The GE F414 engine impasse:** once hailed as the centrepiece of cooperation, the deal remains unresolved; estimated engine cost has nearly tripled to over ₹200 crore, with GE seeking an \$800 million Indian investment for a dedicated production line.
- A pattern of stalled initiatives: the DTTI (2012) faded into irrelevance; iCET's flagship F414 programme has stalled; INDUS-X and talks on Javelin and Stryker co-production have similarly failed to deliver.
- **The technology transfer divide:** India seeks technology and manufacturing know-how for indigenous capability, while the US treats such technologies as strategic assets restricted under its regulations, a structural mismatch.
- Officials see the proposed RDPA as the next test, though critics worry it could expose India's nascent defence manufacturers to direct competition with larger US firms.

India's Fighter Engine Timeline

Year	Development
1961	HF-24 Marut uses imported British Orpheus engines, for lack of a suitable indigenous engine.
1989	DRDO launches the Kaveri engine project.
2000s	Kaveri fails to meet thrust and reliability requirements for Tejas.
2010s	Tejas Mk-1 enters service with imported GE F404 engines.
2023	India-US announced HAL-GE F414 manufacturing and technology transfer in India.
2024-26	Negotiation delays, pricing concerns, and supply issues affect Tejas Mk-2 and AMCA timelines.

Article : India's Biggest Parking Problem Is That It's Free

The Supreme Court's 19 June judgment on the fundamental right to walk highlighted "the violation of the right to walk on demarcated footpaths," drawing attention to footpaths being used as free parking for vehicles.

- Rising car ownership combined with shrinking free parking space has fuelled a

sense of entitlement to park in front of one's house, sometimes escalating to violent disputes.

- **Ahmedabad's Comprehensive Policy (2026), Maharashtra's "No Parking, No Car" policy (2025), and Bengaluru's Parking Policy 2.0** have introduced proof-of-parking and paid on-street parking rules, though enforcement remains inconsistent.
- Urban economist **Donald Shoup** argues free or subsidised parking represents elite capture of public land, worsening inequality and congestion, versus "Right to the City" advocates who see paid parking as disadvantaging the poor.
- Experts broadly favour regulated, priced, permit-based residential parking, reclaiming footpaths for pedestrians.

Right to the City

First developed by French sociologist **Henri Lefebvre** in his 1968 book **Le Droit à la Ville**, the concept is the collective right of all urban residents, not just property owners, to access, use, participate in, and shape the city and enjoy the benefits of urban life without exclusion or discrimination.

- The idea gained prominence in the 1990s through geography and urban planning, later reflected in the New Urban Agenda (Habitat III, 2016) and SDG 11, Sustainable Cities and Communities.

Article : Steady in Turbulence - India Must Build Partnerships in This Age of Global Turmoil

PM Modi's recent visits to Australia and New Zealand marked a shift from largely economic and people-centric ties toward a stronger strategic dimension, consolidating India's position across the Indo-Pacific.

Key Concepts

- **The Indo-Pacific construct:** a geopolitical framing treating the Indian and Pacific Oceans as a single strategic theatre, central to countering China's rising assertiveness and ensuring freedom of navigation.
- **India's Act East Policy:** launched in 2014, evolving from the 1990s "Look East Policy," aiming to deepen India's economic, strategic, and cultural engagement with Southeast Asian and Pacific nations.
- **MAHASAGAR** (Mutual and Holistic Advancement for Security and Growth Across Regions): a newer articulation of India's regional vision, extending the earlier SAGAR doctrine focused on the Indian Ocean.
- **Strategic autonomy:** India's foreign policy principle of engaging multiple powers without becoming formally allied to any one bloc, preserving independent decision-making in foreign and security policy.

Article : Tourism Can Unlock India's Journey to a New Milestone

- India's tourism and hospitality sector contributes about 7% of GDP and accounts for over 9% of total employment, generating multiple indirect jobs for every direct job created.
- The hospitality sector has moved decisively beyond pandemic-era disruption, with occupancy levels and average room rates surpassing pre-pandemic levels and demand continuing to grow.
- Tourism growth is increasingly driven by Tier-II and Tier-III cities rather than a

handful of metropolitan destinations, creating a more diversified tourism ecosystem.

- Expansion of regional airports, road networks, and industrial corridors is reshaping tourism flows; airport-centric development is emerging as a significant driver of hotel growth and business travel.
- **FAITH** calls for granting industry status to tourism nationwide, fast-tracking 50 mission-mode destinations, a liberalised e-visa regime, and a single-window clearance mechanism for private participation.
- **"Industry status"** for tourism means a formal government classification making tourism businesses eligible for benefits available to other industries, priority-sector lending, tax incentives, easier institutional finance, something tourism in most states currently lacks.
- **MICE tourism** (Meetings, Incentives, Conferences and Exhibitions) is a high-value business tourism segment generating significant revenue through corporate travel and events, distinct from leisure tourism.
- Tourism's multiplier effect generates indirect and induced economic activity across transport, retail, food services, and construction, making its total economic impact larger than direct tourism jobs alone suggest.
- Despite significant natural, cultural, and heritage assets, India's global tourism competitiveness has historically lagged behind smaller countries with more developed infrastructure and simpler entry processes.

Article : BSF Initiates Technical Study to Fence the Sundarbans

The Union Home Ministry is set to fence about 90 km of the Sundarbans along the India-Bangladesh border in West Bengal, terrain of mangrove forests, tidal swamps, embankments, and numerous creeks.

- The BSF has initiated a feasibility and technical study to erect the fence; at present there are no barriers on the ground, and Floating Border Outposts (FBOPs) secure the riverine areas.
- This boundary was determined under the **Radcliffe Award**, which divided the erstwhile province of Bengal into two parts.
- The Sundarbans hosts the largest mangrove forest in the world, lying on the delta of the Ganges, Brahmaputra, and Meghna rivers on the Bay of Bengal; 40% lies in India and the rest in Bangladesh.
- At about 4,096 km, the India-Bangladesh border is India's longest land border, passing through West Bengal, Assam, Meghalaya, Tripura, and Mizoram; most of it is already fenced except riverine and marshy stretches like the Sundarbans.

Central Armed Police Forces (CAPF)

The **CAPF** refers to seven security forces under the Ministry of Home Affairs. Each has its own officer cadre but is headed by Indian Police Service officers.

- **Assam Rifles (AR)**: under dual control (operational control under the Army, administrative control under MHA); originated in 1835 as the "Cachar Levy" to protect British tea estates from tribal raids; conducts counter-insurgency in the northeast, secures the Indo-China and Indo-Myanmar borders in peacetime, and provides rear-area security during war.
- **Border Security Force (BSF)**: established 1 December 1965; a border-guarding force charged with guarding India's land borders with Pakistan and Bangladesh.
- **Central Industrial Security Force (CISF)**: established 1969 with 3 battalions to

secure PSUs; the CISF (Amendment) Act, 2009 allowed deployment to private and cooperative establishments on a user-charge basis and authorised protection of Indian missions abroad and UN peacekeeping participation. It now secures Parliament, airports, seaports, metros, nuclear and space installations, and other critical infrastructure.

- **Central Reserve Police Force (CRPF):** began as the Crown Representative's Police on 27 July 1939, becoming the CRPF on enactment of the CRPF Act on 28 December 1949; its mission is to help the government maintain rule of law, public order, and internal security, including crowd and riot control and countering Left-Wing Extremism.
- **Indo-Tibetan Border Police (ITBP):** raised on 24 October 1962; deployed on border-guarding duty from the Karakoram Pass in Ladakh to Jachep La in Arunachal Pradesh, covering 3,488 km of the Indo-China border, alongside internal security roles including anti-Left-Wing-Extremism operations in Chhattisgarh.
- **Sashastra Seema Bal (SSB):** established 15 March 1963 after the 1962 Sino-Indian conflict (previously the Special Service Bureau), created to guard India's borders with Nepal and Bhutan.
- **National Security Guard (NSG):** MHA's nodal national counter-terror force, modelled on the UK's SAS and Germany's GSG-9; comprises the Special Action Group (SAG, drawn from the Army) and the Special Ranger Groups (SRG, drawn from Central Armed Police Forces and State Police).

Article : Kudankulam Nuclear Plant Data Leak Sparks 'Absolute Commotion'

India's Nuclear Power Stations

Plant	Location
Kalpakkam	80 km south of Chennai, on the Coromandel Coast
Kudankulam	Tirunelveli district, Tamil Nadu; under construction with Russian assistance
Kaiga	Uttar Kannada district, Karnataka
Tarapur	India's first nuclear reactor
Kakrapara	Near Surat, on the bank of the Tapi, Gujarat
Rawatbhata	On the bank of the Chambal River, about 65 km from Kota, 3 km from Rana Pratap Sagar
Narora	Bulandshahr district, Uttar Pradesh, on the bank of the Ganga

Article : Gujarat's Data Centre Policy - Green Energy Push, 7.5-GW Capacity Target

A **data centre** is a facility housing computing infrastructure that powers cloud

computing, AI model training, and digital services, increasingly power- and water-intensive as AI adoption grows.

- States compete for data centres because they bring large capital investment and specialised jobs, positioning a state as a digital economy hub, but they also strain local electricity and water resources.
- **"Hyperscale"** refers to data centre facilities built at massive scale, typically for major cloud/AI providers, capable of rapidly scaling compute and storage resources.

Article : Corporate Average Fuel Efficiency (CAFE) Norms

CAFE norms are government-regulated standards mandating automobile manufacturers to meet specific fuel efficiency and emission targets across their entire vehicle fleet. They are implemented under the **Energy Conservation Act, 2001**, and fall under the **Ministry of Power** (through the Bureau of Energy Efficiency).

- Objectives: lower greenhouse gas emissions, especially CO₂, and reduce India's dependence on crude oil imports.
- CAFE norms apply to fleet-wide average emissions, not individual vehicles, and are implemented in phases: **CAFE-1** (introduced 2017), **CAFE-2** (from 2022), and **CAFE-3**, expected from April 2027.
- These norms form part of India's broader climate commitments, including achieving net-zero emissions by 2070.

BS Norms vs CAFE Norms

Feature	Bharat Stage (BS) Norms	CAFE Norms
What they reduce	Toxic tailpipe pollutants like particulate matter (e.g. PM ₁₀ , PM _{2.5})	Carbon dioxide (CO ₂) emissions
How applied	A strict, maximum legal limit on every individual vehicle manufactured	A sales-weighted average across an automaker's entire model lineup
Objective	Improve local air quality and public health	Improve national energy security and reduce fuel consumption
Coverage	All two-wheelers, three-wheelers, and passenger/commercial vehicles	Passenger vehicles up to 3,500 kg (petrol, diesel, CNG, hybrid, EV)

- **BS4 fuel** contains more sulphur (50 parts per million), producing more pollution when burned, and allows nitrogen oxide emissions up to 250 mg/km.
- **BS6 fuel** contains almost no sulphur (only 10 parts per million), making it about 5 times cleaner, and allows nitrogen oxide emissions only up to 80 mg/km.

Article : NITI Aayog's Roadmap for India as a Leading Bioeconomy Powerhouse

About NITI Aayog

- Replaced the Planning Commission on 1 January 2015 as the apex think tank designing India's strategic policy framework.
- Drives cooperative federalism through a bottom-up approach involving all state Chief Ministers in national policymaking.
- Chaired by the Prime Minister, with an appointed Vice Chairperson and CEO; spearheads national growth programmes including the Atal Innovation Mission for technology and entrepreneurship.

About Bioeconomy

The Department of Biotechnology (DBT) defines **bioeconomy** as the economic value generated from the sustainable and innovative use of biological resources (plants, animals, microorganisms, biomass), biological knowledge, and biotechnology-based products and services.

Six National BioMissions

Mission	Focus
GeneIndia	Affordable gene and cell therapies, better treatments
AgriBio 2.0	Climate-resilient gene-edited crops, biological farm inputs
BioX Foundry	Uses microorganisms like bacteria to make useful products, e.g. eco-friendly plastics
One Health Grid	Tracks diseases in humans, animals, and the environment to stop outbreaks early
Marine Biotechnology	Benefits from the sea, e.g. seaweed cultivation
BioPharmaNext	Better medicines and vaccine production

Key Features of the Roadmap

- A ₹50,000 crore fund to support new science and biotechnology projects.
- **Production-Linked Incentives (PLI)** rewarding companies that manufacture biotechnology products in India.
- Support for better medicines and disease testing, and industry-academia collaboration to solve real-life problems.
- Aims to make new-medicine approvals faster, slashing the current 900-day average approval time down to competitive global standards.

Article : Sleepless Nights, A Public Health Challenge

A US-based climate advocacy group assessed sleep loss across over 1,300 cities worldwide, finding the average person lost nearly 56 hours of sleep annually between

2020 and 2025 due to high night-time temperatures; at least 10% of this loss is directly attributable to climate change.

City-wise Data (India)

- **Chennai:** 93 hours lost; **Mumbai:** 84 hours; **Delhi:** 67 hours, all above the global average.

Underlying Drivers and Health Stakes

- Longer, hotter, more humid summers intensified by the urban heat island effect, compounded by screen time, social-media use, erratic eating habits, and shift-based work.
- Sleep is essential for cellular repair, immune function, and regulating inflammation; chronic sleep deprivation is linked to hypertension, diabetes, heart disease, obesity, and cognitive impairment.

What Is Sleep Health

- Adequate, quality sleep (7-9 hours for adults) is essential for physiological and cognitive restoration; sleep cycles (REM and Non-REM) regulate memory consolidation, metabolism, and immunity.

Urban Heat Island (UHI) Effect

- Concrete surfaces, reduced vegetation, and human activity make cities warmer than surrounding areas, especially at night; persistently warm nights ("tropical nights") prevent the body's natural cooling needed for sleep onset.

Heat Action Plans (HAPs) in India

- First introduced in Ahmedabad (2013) after the 2010 heatwave deaths, now adopted by multiple states under NDMA guidance; current plans focus mostly on daytime advisories, water stations, and early warning systems.

Link to Non-Communicable Diseases (NCDs)

- India faces a rising burden of NCDs (diabetes, hypertension, cardiovascular disease); the WHO recognises inadequate sleep as a contributing behavioural risk factor for NCDs.

Article : SC Urges Railways to Look Into Overcrowded Trains

A bench headed by Justice Sanjay Karol flagged the persistent menace of overcrowding on trains, urging the Railways to deploy personnel on the ground to stop people travelling on footboards.

- The Court held that passengers share equal responsibility, describing train-travelling "daredevils" as unwilling to reform.
- It asked the Railways to reconsider the "second-class" nomenclature as an anachronism alien to the Constitution's egalitarian spirit.
- The case arose from an appeal by Lata, whose husband died in November 2015 after falling from a moving train; she had earlier been denied compensation by the Railway Claims Tribunal and the Madhya Pradesh High Court for want of a recovered ticket.

Article : India's Funds, Talent, and the Opening for a Research Leap

India spends 0.64% of GDP on R&D (2023), against a global average more than twice

that; China spends 2.43%. Only about two-fifths of India's R&D is financed by private enterprise, versus three-quarters or more in China, Korea, and the US.

- What's required next: industry must mobilise its own capital behind the public catalyst; a strategic choice of technologies where import-dependence is most dangerous and capability most valuable; and institutional machinery within firms, dedicated research units, and corporate venture arms insulated from quarterly pressures.

Anusandhan National Research Foundation (ANRF)

- Established through the **Anusandhan National Research Foundation Act, 2023**, functioning under the Department of Science & Technology (DST).
- The Science and Engineering Research Board (SERB) has been merged into ANRF.
- Acts as an apex body providing high-level strategic direction for scientific research, per National Education Policy recommendations.
- **Funding target:** aims to mobilise ₹50,000 crore during 2023-28 through multiple streams, including the ANRF Fund, Innovation Fund, Science and Engineering Research Fund, and Special Purpose Funds.
- Prepares the roadmap for short, medium, and long-term R&D, and seeds and grows research at academic institutions, particularly universities and colleges with nascent research capacity, through R&D projects, fellowships, academic chairs, and centres of excellence.

Article : Battling Drug Abuse - Kerala's Coordinated Anti-Cartel Push

NDPS cases in Kerala surged from 5,695 in 2021 to 26,619 in 2022, rising further to 36,314 in 2025, alongside large seizures. Ernakulam city accounts for a substantial share of cases; synthetic drug cartels use digital platforms and social media to outpace enforcement.

Operation Toofan

- Launched by Kerala's UDF government in June, integrating the State police, southern-state police forces, central agencies, and the education, health, and excise departments.
- Until 15 July, the operation had netted over 7,600 drug peddlers in about 7,100 cases, with synthetic drugs forming a significant share of seizures.
- Coordination is being strengthened under the **NCORD (Narco-Coordination)** framework, including the NIDAAN database of arrested narco-offenders; Kerala's Home Minister is meeting other Chief Ministers to build inter-state coordination, with nodal officers being assigned.

Institutional Framework

- **NDPS Act, 1985:** India's primary law criminalising production, possession, sale, and trafficking of narcotic and psychotropic substances.
- **NCORD Mechanism:** a multi-tier mechanism (Centre, State, District) set up by the MHA for inter-agency coordination on drug enforcement.
- **Narcotics Control Bureau (NCB):** the nodal drug law enforcement and intelligence agency under the MHA, constituted on 14 November 1985 under the NDPS Act.

Article : The Delimitation Debate Is a Fight Over the Wrong Number

The author argues the delimitation debate wrongly fixates on population alone as the

sole basis for parliamentary seat-share.

- Southern states (Tamil Nadu, Karnataka, Telangana) contribute roughly 20% of all taxes, while northern states (Uttar Pradesh, Bihar, Madhya Pradesh) contribute about 6%, yet the southern population share is falling while the northern share is rising.
- Southern states' population share is projected to fall from about 13% to 12%, while northern states' share is projected to rise from about 32% to 34%, over a ten-year window.
- The author calls for the Delimitation Commission to weigh a composite formula of population and economic contribution, akin to how the Finance Commission allocates devolution.
- A purely population-based formula has already prompted proposals to incentivise higher birth rates in southern states to protect future seat share, an incentive the author flags as warped.

Article : India Rejects Former Japan Minister's Remarks on Bullet Train Delay

The **Mumbai-Ahmedabad High-Speed Rail (MAHSR)** is India's first bullet train project, a 508 km corridor using Japanese Shinkansen technology under a 2015 India-Japan agreement.

- Funded largely through a low-interest soft loan from the Japan International Cooperation Agency (JICA), implemented by the National High Speed Rail Corporation Limited (NHSRCL).
- **Shinkansen** is Japan's high-speed rail system, in operation since 1964, renowned for its speed, safety record, and precision.
- BEML and the Integral Coach Factory (ICF) are developing India's own high-speed trainsets under "Make in India," reducing dependence on imported rolling stock for initial operations.
- MAHSR is a flagship project under the India-Japan Special Strategic and Global Partnership, symbolising technology transfer and infrastructure cooperation.

Article : Non-Animal Testing Models (NAMs) in Drug Development

Non-Animal Testing Models (NAMs) are emerging as scientifically credible alternatives and complements to traditional animal testing, presenting an opportunity for India to strengthen its pharmaceutical innovation ecosystem.

NAMs are scientific approaches using human-relevant biological systems and computational methods to generate safety, efficacy, and toxicity information, instead of relying solely on animal testing.

Animals Traditionally Used in Drug Testing

- **Caenorhabditis elegans (C. elegans)**: a tiny, roughly 1mm worm with a see-through body, letting scientists watch its cells work in real time; used to study aging and genes given its short life.
- **Drosophila melanogaster**: the common fruit fly, breeding a new generation every 10 days; about 60% of its genes are similar to human genes linked to disease.
- **Zebrafish**: a small, striped freshwater fish whose embryos are transparent, letting scientists watch organs form; it shares about 70% of its genes with humans.

Stem Cells and Organoids

Stem cells are special cells that can grow into many different types of body cells, such as liver, brain, or heart cells. Scientists grow them into **organoids**, tiny, mini versions of real body organs, like a mini-liver or mini-brain, inside a dish.

- Because organoids behave like small working organs, they can be used to see how a new medicine affects the body, removing the need to test on animals; this is the core idea behind NAMs.

Types of NAM Platforms

- **Organoids:** miniature 3D structures grown from human stem cells that mimic the architecture and function of human organs.
- **Organ-on-chip systems:** microfluidic devices that recreate the physiological environment of human tissues and organ interactions.
- **Three-dimensional human cell culture models:** advanced cellular systems that better replicate human tissue behaviour than conventional 2D cultures.
- **Computational toxicology:** computer-based models predicting chemical toxicity and drug responses using biological and chemical data.
- **AI-enabled models:** machine learning systems analysing large biomedical datasets to predict human drug responses.

Generic Drugs vs Biosimilars

- **Generic drug:** a copy of a normal chemical medicine after its patent ends; since it is a simple chemical, it can be copied exactly, like photocopying a document.
- **Biosimilar:** a copy of a biologic medicine (made from living cells, not chemicals); since living systems are complex, it can only be "highly similar," not identical, to the original.

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